

Standard Model Particles as Closed Walks on K_7 , I: Torus-Knot Carriers, the Steane $[[7, 1, 3]]$ Code, and a Closed-Form Mass Spectrum (Paper 21a — theory companion)

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Abstract

The Null Worldtube (NWT) program derives Standard Model structure from the algebra $K_7/\mathfrak{so}(7)/\text{Spin}(7)/\text{Cl}(0, 7)$. This is the theory companion to the experimental Paper 21b; it assembles the load-bearing theoretical results that turn that algebra into two things: a closed-form derivation of the Standard Model fermion and hadron mass spectrum, and a directly-executable protocol that realizes each particle’s walk as a state on 7-qubit Steane $[[7, 1, 3]]$ stabilizer-code hardware. The central identifications are: (i) every Standard Model fermion and hadron is a closed walk on the Heffter embedding of K_7 , with the walk’s (p, q) torus-homology class labeling the Paper 6/7 quantum numbers (mass, charge, isospin, and baryon number); (ii) the carrier-knot family entering the mass formula is the $(2, F_n)$ -torus-knot family {unknot, Hopf, trefoil, cinquefoil, $T(2, 8), T(2, 13), \dots$ } for consecutive Fibonacci F_n , with the family forced by K_7 ’s binary Hamilton bifurcation ($d = 1$ QR, $d = 3$ NR; $d = 2$ substrate-forbidden) and the Fibonacci indexing forced by Φ -shell substrate algebra; (iii) the closed-form mass-formula factor is $n_q = \det(K) = F_n$ via the Murasugi formula and $n_q^q = \det(K^{\sqcup q}) = \dim((\text{spin}-((n_q - 1)/2))^{\otimes q})$ via determinant multiplicativity / $SU(2)$ tensor power; (iv) per-walk sector assignment $n_q = f(p \bmod 7, q \bmod 7)$ follows the cosmogenic $Z_3 = \text{AGL}(1, 7)$ generator $x \mapsto 2x \bmod 7$, the same Z_3 that gives three fermion generations; (v) the Paley quadratic-residue decomposition of K_7 is the Steane $[[7, 1, 3]]$ stabilizer-and-codespace decomposition, with the walk-to-Pauli direction-sensitive map sending QR-direction steps to logical X and NR-direction steps to logical Z ; (vi) the transversal Hadamard $H^{\otimes 7}$ is the substrate’s matter–antimatter CPT operation, and pair production proceeds by simultaneous reverse-traversal walk emission from substrate vacuum; (vii) the syndrome classification of the 25-particle compendium reduces to 10 distinct (s_X, s_Z) pairs with falsifiable cross-sector substrate-equivalences (e.g. $\mu^- \equiv D^+ \equiv J/\psi$ at the syndrome level); (viii) the substrate predicts a catalog of sub-500 MeV unseen species, a $q \equiv 6 \bmod 7$ heavy-species class, and a Fibonacci-spaced light-particle tower at MeV scale. The closed-walk framework applies to matter content; bosons enter as substrate-mediator operations, not walk excitations, and the hardware-realization claim of Paper 21b is restricted to fermions and hadrons accordingly. No free parameters enter the derivation: the chain $(|p|, |q|) \rightarrow \text{sector} \rightarrow \text{carrier}(2, F_n) \rightarrow n_q^q$ is closed-form from K_7 substrate primitives (binary Hamilton bifurcation, Φ -shells, cosmogenic Z_3 , Murasugi formula, $SU(2)$ tensor power, Paley QR/NR partition).

1 Introduction

The Null Worldtube (NWT) framework [9, 13, 14, 15, 16] is a substrate-monism research program in which the fine-structure constant [9], Newton’s constant [13], the full Standard Model spectrum [6,

11], three fermion generations [16], the neutrino sector [16], and seven cosmological parameters [20] are derived from a single substrate algebra $K_7/\mathfrak{so}(7)/\text{Spin}(7)/\text{Cl}(0,7)$, with no free *dimensionless* parameters: the only dimensional inputs are $\hbar$ (a unit-conversion factor), the cosmogenic mass scale M_{Pl} , and the realization's substrate primitives (a, J_{int}) that fix c (§9.5).

This paper consolidates the theoretical chain by which the substrate algebra is reduced to a directly-executable protocol on quantum hardware. It is the theory companion to Paper 21b [17], which reports the bench-scale experimental confirmation; this paper provides the substrate-mathematical derivation.

The closure status that motivates this companion paper. Paper 6's mass formula [6] and Paper 8's carrier-knot table [8] reproduce the 80-particle Standard Model spectrum at median $\sim 0.1\%$ PDG precision [11], from two inputs (the electron mass scale m_e and a single dimensionless gauge normalization κ_{SM}) plus the proton- and Z -mass anchors (§9.2, Table 4). The persistent fitting critique against that program was that the carrier-knot crossing-number $n_q \in \{0, 2, 3, 5\}$ entering the formula was *empirically* assigned per sector rather than *substrate-canonically* derived from K_7 . A sequence of closure attempts via $\mathfrak{so}(7)$ and $\text{Spin}(7)$ representation theory produced negative results: no single $SU(2) \subset \text{Spin}(7)$ on the $\text{Cl}(0,7)$ spinor space $S = 8$ realizes the required dimension 5 irreducible with the right walk projection, and Wilson-loop correlations against n_q plateau at $r \lesssim 0.6$. The closure presented here uses a different route: knot-theoretic identification of the carrier family as $(2, F_n)$ -torus knots for Fibonacci index F_n , closed-form n_q via Murasugi's formula [4], closed-form n_q^q via $SU(2)$ tensor power, and per-walk sector assignment via the cosmogenic $Z_3 = \text{AGL}(1,7)$ generator. The result is end-to-end closed-form: no free parameters, no fitting, no representation-theoretic input beyond elementary $SU(2)$ tensor decomposition.

Outline. §2 reviews the NWT substrate algebra. §3 develops the closed-walk framework: Heffter embedding, (p, q) torus-homology, walk enumeration via the Bogoliubov pipeline, Hamilton-cycle taxonomy, the Paley $(7, 3, 1)$ Fano structure, the walk-knot vs carrier-knot distinction, and the scope restriction to fermions and hadrons. §4 develops the carrier-knot family closure: the $(2, F_n)$ torus-knot family, closed-form n_q via Murasugi, closed-form n_q^q via $SU(2)$ tensor power, and the cosmogenic- Z_3 sector rule. §5 develops the $K_7 = \text{Steane } [[7, 1, 3]]$ identification: Paley stabilizers, walk-to-Pauli direction-sensitive map, transversal Hadamard CPT. §6 presents the substrate-vacuum pair-production mechanism with worked examples for $e^\pm$ and $p\bar{p}$. §7 catalogs the syndrome classification of the 25-particle compendium and its substrate-canonical cross-sector equivalences. §8 compiles the substrate's falsifiable predictions: missing (p, q) species, $q \equiv 6 \pmod 7$ classes, and the Fibonacci-spaced light-particle tower. §9 closes with the fitting-critique status and the bridge to Paper 21b's experimental confirmation.

2 NWT substrate primer

The NWT framework [9, 13, 14, 15, 16] identifies fundamental physics with the algebra $K_7/\mathfrak{so}(7)/\text{Spin}(7)/\text{Cl}(0,7)$, where:

- K_7 is the complete graph on 7 vertices, embedded on a torus T^2 via the Heffter embedding (vertex k at $(k/7, 3k \pmod{7/7})$);
- $\mathfrak{so}(7)$ is the 21-dimensional Lie algebra acting on $\mathbb{R}^7$, the imaginary part of the octonions $\mathbb{O}$;
- $\text{Spin}(7)$ is the simply-connected cover of $SO(7)$;
- $\text{Cl}(0,7)$ is the Clifford algebra of $\mathbb{R}^{0,7}$, with spinor space $S = 8$.

The matter content treated in this paper lives on K_7 ; its K_8 extension—the complete graph on 8 vertices, equivalently the $[[8, 1, 1]]$ stabilizer code—carries the dark and neutrino sectors through Spin(8) triality [16] and is treated here only in outline (§8.5, §9.4).

The substrate-derived quantities established in prior NWT papers include the fine-structure constant [9]

$$\alpha = \frac{1}{25\pi\sqrt{3} + 1} \approx 7.297 \times 10^{-3} \quad (1)$$

(a zero-parameter closed form accurate to 7.6 ppm: $\alpha^{-1} = 25\pi\sqrt{3} + 1 = 137.0350$ versus the CODATA 137.035999; 7.6 ppm is the stated accuracy of the trefoil expression [9], with the residual an uncomputed higher-order correction—the program does *not* claim sub-ppm α), the gravitational-constant chain [13], the full 80-particle Standard Model spectrum at median $\sim 0.1\%$ PDG precision [11], three fermion generations from $G_2 \rightarrow SU(3)$ breaking via S^6 stabilizer direction [16], the neutrino sector [16], and the cosmological-constant cascade $\Lambda_{\text{cc}} = (m_e/M_{\text{Pl}})^4 \cdot \alpha^{16} \cdot h_{\text{Coxeter}}$ at three-digit Planck precision [20].

For this paper, the load-bearing substrate primitives are:

- $h_v = 5$ (Hopf vacuum, the closure-2 substrate vertex count);
- $\text{rank}(\mathfrak{so}(7)) = 3$ (rank of $\mathfrak{so}(7)$, three fermion generations);
- $\dim(G_2) = 14$ (dimension of $G_2 = \text{Aut}(\mathbb{O})$, the octonion automorphism group);
- α (the trefoil Aharonov–Bohm phase coupling);
- the Paley $(7, 3, 1)$ design on K_7 , with quadratic residues $\text{QR}_7 = \{1, 2, 4\}$ and non-residues $\text{NR}_7 = \{3, 5, 6\}$;
- the cosmogenic $Z_3 = \text{AGL}(1, 7)$ generator $x \mapsto 2x \bmod 7$;
- the Φ -shell Fibonacci/Lucas substrate ladder from prior NWT closure work.

Each of these primitives is used below to derive specific load-bearing identifications.

3 Particles as K_7 closed walks

3.1 Heffter embedding and the (p, q) torus-homology classification

The Heffter embedding of K_7 on the torus [3] places vertex k at $(k/7, 3k \bmod 7/7) \in T^2$, with edges drawn as geodesic chords on the flat torus. Each closed walk $W = v_0 \rightarrow v_1 \rightarrow \cdots \rightarrow v_L \rightarrow v_0$ on K_7 inherits a homology class $(p, q) \in H_1(T^2; \mathbb{Z})$ from its lifted trajectory on the universal cover $\mathbb{R}^2$, where p counts net u -direction windings and q counts net v -direction windings.

The substrate identification of Standard Model fermions and hadrons [6] maps each particle to a closed walk on K_7 whose (p, q) windings label the Paper 6 torus-knot mass-formula quantum numbers. The walk’s (p, q) class is invariant under reparametrisation of the walk’s start vertex but is sensitive to reversal: $\bar{W}$ = reverse traversal of W has homology class $(-p, -q)$ (see §6).

3.2 Bogoliubov-pipeline walk enumeration

The walk-as-particle correspondence was originally posited per-particle on heuristic grounds in Paper 6. Subsequent computational work in the Bogoliubov-pipeline phase-E–H arc [21] replaced the per-particle posit with a systematic enumeration: starting from the polar vertex $P = 0$, all closed walks of length $L \leq 25$ on K_7 are enumerated by breadth-first search; walks with the same $(|p|, |q|)$

class are grouped; and the shortest-length representative per class is identified. The result is a complete walk classification covering the 25-particle Standard Model compendium (Table 1, with all walks listed in [21]), plus a catalog of 13 walk-realizable $(|p|, |q|)$ classes *not* in the SM compendium (§8).

Table 1: Compendium of the 25-particle Standard Model walks on K_7 via Bogoliubov-pipeline enumeration [21]. $L_{\min}$ is the shortest-length walk in the $(|p|, |q|)$ class. All walks start and close at the polar vertex $P = 0$.

(p , q)	Sector	Particles in class	$L_{\min}$
(1, 3)	nucleon	$p, n, \Sigma^+, \Sigma^0, \Sigma^-$	7
(2, 1)	lepton	e^-	5
(3, 5)	meson	π^+	8
(1, 8)	lepton	μ^-	22
(2, 5)	meson	K^+	10
(2, 7)	meson	$D^+, J/\psi$	14
(3, 4)	hyperon	$\tau^-, \Lambda, \Sigma^*$	11
(3, 7)	meson	D^0	17
(4, 5)	meson	ω	13
(4, 9)	meson	Υ	27
(5, 4)	hyperon	Ξ^0, Ξ^-, Δ	13
(5, 7)	meson	ρ	18
(6, 5)	meson	η	17
(7, 3)	meson	π^0	10
(7, 4)	hyperon	Ω^-	12
(7, 5)	meson	K^0	11

The enumeration discloses the structural decomposition of long walks into shorter sub-walks: μ^- at (1, 8) realizes as five concatenated (0, 1) tri-cycles plus the K_7 Hamilton cycle; Υ at (4, 9) realizes as three concatenated (1, 2) pent-cycles plus the Hamilton cycle; and so on [21]. These sub-cycle decompositions become structurally important in §3.6 (substrate-mediator interpretation of bosons) and §8 (substrate-prediction catalog).

3.3 Three Hamilton-cycle types by edge-difference

The complete graph K_7 admits three combinatorially distinct Hamilton cycles, classified by signed edge-difference $d \pmod{7}$:

$$d = 1 \text{ (QR)} : 0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 0 \quad (\text{proton walk}) \quad (2)$$

$$d = 2 \text{ (QR)} : 0 \rightarrow 2 \rightarrow 4 \rightarrow 6 \rightarrow 1 \rightarrow 3 \rightarrow 5 \rightarrow 0 \quad (3)$$

$$d = 3 \text{ (NR)} : 0 \rightarrow 3 \rightarrow 6 \rightarrow 2 \rightarrow 5 \rightarrow 1 \rightarrow 4 \rightarrow 0 \quad (\text{NR-Hamilton}) \quad (4)$$

The $d = 4, 5, 6$ cycles are reverses of $d = 3, 2, 1$ respectively. The walk’s d -direction Hamilton-cycle content determines the carrier-knot sector [23]: walks dominated by $d = 1$ (QR) become nucleons, walks dominated by $d = 3$ (NR) become hyperons or CP-coupled species, mixed-Hamilton walks become mesons, walks with no full Hamilton become leptons.

Crucially, the $d = 2$ Hamilton cycle is *unrealised in any Standard Model particle*: every walk in the 25-particle compendium has $\text{ham}_{d=2} = 0$. This is a substrate-canonical absence: the $d = 2$ “skip-one” tetrahedral direction does not realize stable bound-state closures, even though it is combinatorially valid. The K_7 binary Hamilton bifurcation $\{d = 1, d = 3\}$ becomes load-bearing in §4.1 for the “2” in the carrier-knot family $(2, F_n)$.

3.4 Paley (7, 3, 1) Fano structure and Hamilton-containment law

The Paley (7, 3, 1) design on K_7 [1] provides seven Fano triangles, namely the QR translates of $QR_7 = \{1, 2, 4\}$:

$$F_i = \{i + 1, i + 2, i + 4\} \pmod{7}, \quad i = 0, \dots, 6. \quad (5)$$

This Steiner triple system covers each K_7 edge in exactly one Fano triangle. The remaining $35 - 7 = 28$ non-Fano triangles cover each K_7 edge in exactly four.

Hamilton-containment law. A K_7 closed walk contains a Hamilton cycle as a sub-walk if and only if it has full Fano-edge coverage (touches all seven Fano triangles via edges, not merely vertices). This is an empirical regularity across the 16 compendium (p, q) classes (not yet a proven theorem) [23]: 13 of 16 have full Fano coverage, and the three exceptions (e^- , Λ family, π^+) are exactly the walks that visit fewer than seven K_7 vertices. The structural reading: the Hamilton scaffold is the substrate manifestation of the Steiner (7, 3, 1) covering. The regularity sharpens to a candidate theorem—full Fano-edge coverage $\Leftrightarrow$ Hamilton-cycle containment, the three exceptions being exactly the sub-Hamilton walks (those visiting fewer than seven vertices)—whose proof at the level of K_7 walk combinatorics remains open.

3.5 Walk-knot versus carrier-knot

A subtle distinction operative throughout this paper: the (p, q) torus knot of the walk’s three-dimensional Heffter embedding is *not* the same as the Paper 8 carrier knot. The Paper 8 carrier-knots (unknot, Hopf link, trefoil 3_1 , cinquefoil 5_1 , plus substrate-prediction extensions) are determined by the walk’s σ -orbit transition structure and by the substrate’s binary Hamilton bifurcation plus Φ -shell algebra, *not* by spatial embedding. Direct computation confirms this dichotomy operationally: lifting each compendium walk to a three-dimensional Heffter-torus embedding and reading off the embedded-knot determinant via `pyknotid` gives mostly the unknot, with at most 14% agreement with the Paper 8 carrier-knot. The carrier-knot identification is at the *substrate algebra* level, not the three-dimensional embedding level. To commit on what fixes the carrier: there is *no* direct walk-to-knot map through the embedding (the `pyknotid` lift gives mostly unknots); the carrier is *defined* by the substrate-algebra assignment of §4—the $(2, F_n)$ family selected by the σ -orbit and Φ -shell structure via rule (13)—and its load-bearing content is the integer $n_q = F_n$ and the associated $SU(2)$ representation, with “carrier knot” a geometric label for that algebraic object.

3.6 Scope: fermions and hadrons, not bosons

The closed-walk framework above describes the *charged matter content*: charged leptons, mesons, hyperons, and nucleons. Each is a closed walk on K_7 with definite (p, q) winding, Hamilton-cycle content, and carrier-knot assignment. The closed-walk framework does *not* extend to (i) the bosonic gauge sector—photons, gluons, $W^\pm$, Z , and the Higgs are not represented as closed K_7 walks of definite (p, q) winding—nor to (ii) neutrinos, which require the K_8 Wilson-amplitude extension of Paper 20 [16] rather than a K_7 closed-walk carrier (see paragraph below on the neutrino sector).

Bosons enter NWT not as walk-as-particle excitations but as *substrate-mediator operations* acting on the matter Hilbert space, and the hardware-realization claim of Paper 21b [17] is restricted accordingly to fermions and hadrons. Two complementary, dual readings frame the boson-sector algebra [22]. In the *single-patch* reading a boson is a logical-Pauli operator in $N(S_N)/S_N$ on a multi-patch Steane code: the $U(1)$ photon is the polar-vertex pair $(X_P^{(i)} X_P^{(j)}, Z_P^{(i)} Z_P^{(j)})$, its spin-1 emerging from the CSS X/Z duality, and the gluon ($\mathfrak{su}(3)$), the $W^\pm$ weak vertex, and the Higgs ($g^{\text{Higgs}} = Y_0 Y_7$ on the K_8 extension) follow as the closures summarized in §9.4. In the dual *multi-patch* reading a boson is the inter-patch entanglement (CNOT-bridge) channel between two fermion walks, which gives scattering

a direct chip realization (prepare two walks, entangle through the mediator channel, read joint syndromes). Pair production (§6) is the substrate-vacuum mediator operation that emits a matter walk and its reverse-traversal antimatter partner simultaneously, triggered at the polar vertex; scattering, decay, and nuclear-binding vertices are likewise mediator operations rather than walks (§9.4). On chip these are realized as gate sequences— $H^{\otimes 7}$ for CPT, stabilizer measurements for gauge readout, CNOT-bridges for inter-walk mediation—not as encoded boson excitations.

Neutrinos: K_8 Wilson amplitude, not K_7 closed walks. Neutrinos are fermions but their substrate realization lies outside the closed-walk + Paper 6 torus-knot framework of this paper. The $(2, F_n)$ carrier family on K_7 has no slot below the charged-lepton unknot: charged leptons already occupy the unknot via $n_q = \det(T(2, F_2)) = F_2 = 1$, and the convention $F_1 = F_2 = 1$ leaves no room for a distinct “lighter” unknot carrier within K_7 . The correct substrate framing, established in Paper 20 [16], places neutrinos on the K_8 extension of the substrate graph, with the eighth vertex identified as the real octonion unit $1 \in \mathbb{R} \subset \mathbb{O}$ forced by the $\text{Spin}(7) \subset \text{Spin}(8)$ stabilizer chain. The unified Wilson-amplitude recipe [16, eq. 5.1],

$$m = \frac{8}{N_v} \alpha^{N_e/2} \left(1 + \frac{\alpha}{7}\right) (1 + 3\alpha^2) m_{\text{Pl}}, \quad (6)$$

reproduces m_e at 8-significant-figure CODATA precision on K_7 ($N_v = 7$, $N_e = 21$) and gives the lightest active neutrino at $m_1 = 14.84$ meV on K_8 ($N_v = 8$, $N_e = 28$), with $m_2 = 17.16$ meV and $m_3 \approx 53$ meV via NuFIT mass-squared splittings and $\Sigma m_\nu \approx 85$ meV—below the Planck bound (120 meV) but in tension with tightening DESI constraints ($\lesssim 60$ – 72 meV), which makes Σm_ν itself a near-term test of the K_8 mass formula. Neutrinos are therefore handled by Paper 20’s K_8 extension; the closed-walk + carrier-knot machinery of this paper applies to charged leptons, mesons, hyperons, and nucleons. Bench-scale chip realization of neutrinos would require an 8-qubit stabilizer-code patch (e.g. Reed–Muller [8, 3, 2] or a $\text{Spin}(7) \subset \text{Spin}(8)$ -adapted custom code) alongside the 7-qubit Steane patches used for charged matter, and is deferred to a future Paper 21 follow-up.

Throughout the rest of this paper, “particle” means “fermion or hadron” unless otherwise stated.

4 The $(2, F_n)$ carrier-knot family and closed-form n_q^q

This section closes the derivation chain $(|p|, |q|) \rightarrow \text{sector} \rightarrow \text{carrier} \rightarrow n_q^q$ from substrate primitives. The Paper 6 mass formula [6] requires the carrier crossing-number n_q and its q -th power n_q^q . Until 2026-Q2, the values $n_q \in \{0, 2, 3, 5\}$ entered the formula empirically per sector; this section derives them in full closed form. The derived (carrier-determinant) values are $\{1, 2, 3, 5\} = \{F_2, F_3, F_4, F_5\}$: they coincide with the empirical set on the meson, hyperon, and nucleon sectors and differ only for leptons, where the substrate-canonical unknot determinant $n_q = \det T(2, F_2) = F_2 = 1$ supersedes the constituent-count label 0 of Paper 6’s enumeration. Throughout this paper n_q denotes the carrier determinant F_n , not Paper 6’s constituent count.

4.1 K_7 binary Hamilton bifurcation $\rightarrow$ “2” in $(2, F_n)$

The $(2, F_n)$ torus-knot family is the $p = 2$ slice of the two-parameter torus-knot lattice $T(p, q)$. The substrate origin of the $p = 2$ restriction is K_7 ’s binary Hamilton bifurcation (§3.3): only the QR direction $d = 1$ and the NR direction $d = 3$ host realized Hamilton cycles in any Standard Model walk. Empirically verified across the 25-particle compendium: $\text{ham}_{d=2} = 0$ for every compendium walk; the $d = 2$ “skip-one” Hamilton direction is substrate-forbidden. The remaining substrate Hamilton directions form a Z_2 partition $\{\text{QR}, \text{NR}\}$, and the $(2, n)$ -torus-knot family in Schubert 2-bridge classification [5] is the natural geometric realization of that Z_2 .

4.2 Φ -shell substrate algebra $\rightarrow$ Fibonacci indexing

The substrate's Φ -shell algebra (developed in the NWT integer-ladder closure arc) decomposes the substrate vertex/orbit accounting into a Fibonacci/Lucas double ladder governed by golden-ratio shells. Specifically:

- The Lucas half $\{L_2, L_4, L_6, L_8, \dots\} = \{3, 7, 18, 47, \dots\}$ governs the gauge-shell vertex counts (e.g. $L_4 = 7 = |K_7|$).
- The Fibonacci half $\{F_2, F_3, F_4, F_5, F_6, F_7, \dots\} = \{1, 2, 3, 5, 8, 13, \dots\}$ governs the carrier-knot crossing-number sequence.
- The two halves are dual under the golden-ratio recursion $\phi^2 = \phi + 1$, with $L_n = \phi^n + (1 - \phi)^n$ and $F_n = (\phi^n - (1 - \phi)^n)/\sqrt{5}$.

The four Paper 8 carriers $\{\text{unknot}, \text{Hopf}, \text{trefoil}, \text{cinquefoil}\}$ with $n_q \in \{1, 2, 3, 5\}$ are the consecutive Fibonacci subsequence $\{F_2, F_3, F_4, F_5\}$ (the Hopf member is $T(2, 2)$, a 2-component link rather than a knot; “torus-knot family” is shorthand throughout for the torus-knot/link family $T(2, F_n)$). The extension to $F_6 = 8, F_7 = 13, F_8 = 21, F_9 = 34, F_{10} = 55$ provides substrate-prediction carriers for heavier hypothetical species (§8).

4.3 n_q via the Murasugi knot-determinant formula

Combining the binary Hamilton bifurcation (§4.1) and the Fibonacci indexing (§4.2), the substrate carrier-knot family is the $(2, F_n)$ -torus-knot family $T(2, F_n)$ for F_n ranging over consecutive Fibonacci values. The Jones polynomial of the (p, q) -torus knot (with $\gcd(p, q) = 1$) is given in closed form by the Murasugi formula [4]:

$$V(T(p, q))(t) = t^{(p-1)(q-1)/2} \cdot \frac{1 - t^{p+1} - t^{q+1} + t^{p+q}}{1 - t^2}. \quad (7)$$

Evaluating at $t = -1$ gives the knot determinant $\det(K) = |V_K(-1)|$. For the $(2, F_n)$ family this evaluates to F_n exactly:

$$\boxed{n_q = \det(T(2, F_n)) = |V_{T(2, F_n)}(-1)| = F_n.} \quad (8)$$

The compendium verification is exact (Table 2). For non-trivial carriers the Jones polynomial span coincides with the determinant ($\text{span}(V_{T(2, F_n)}) = F_n$ for $n \geq 3$), providing a second closed-form invariant agreeing with n_q .

Table 2: Compendium carriers, their Schubert and torus-knot labels, Jones polynomials, and determinant $= n_q = F_n$ identity. Substrate-prediction extensions to $F_6 = 8$ and $F_7 = 13$ are verified at the polynomial level.

Sector (n_q)	Knot/link	Schubert	Torus	Jones $V_K(t)$	$\det(K) = F_n$
lepton (0/1)	unknot 0_1	$K(1, 0)$	$T(2, 1)$	$+1$	$1 = F_2$
meson (2)	Hopf ⁺	$K(2, 1)$	$T(2, 2)$	$-t^{-5/2} - t^{-1/2}$	$2 = F_3$
hyperon (3)	trefoil 3_1	$K(3, 1)$	$T(2, 3)$	$+t + t^3 - t^4$	$3 = F_4$
nucleon (5)	cinquefoil 5_1	$K(5, 1)$	$T(2, 5)$	$+t^2 + t^4 - t^5 + t^6 - t^7$	$5 = F_5$
<i>prediction</i>	$T(2, 8)$ link	—	$T(2, 8)$	$V(t = -1) = 8$	$8 = F_6$
<i>prediction</i>	$T(2, 13)$ knot	—	$T(2, 13)$	degree-13 alternating	$13 = F_7$

4.4 n_q^q via $SU(2)$ tensor power and disjoint-union determinant

The Paper 6 mass formula uses n_q^q , the q -th power of the carrier crossing-number. Two closed-form identifications of n_q^q are verified across all 8 compendium (n_q, q) pairs.

Identification 1 ($SU(2)$ tensor power). For each carrier $T(2, F_n)$ assign the $SU(2)$ irreducible representation $\rho_n = \text{spin}-((F_n - 1)/2)$ of dimension $F_n = 2j + 1$ (so $\rho_2 = \mathbf{1}$, $\rho_3 = \mathbf{2}$, $\rho_4 = \mathbf{3}$, $\rho_5 = \mathbf{5}$). Then

$$\dim(\rho_n^{\otimes q}) = F_n^q = n_q^q \quad (9)$$

by elementary $SU(2)$ tensor decomposition (Clebsch–Gordan). The substrate reading is that the walk’s q v -cycle revolutions correspond to q parallel carrier-representation copies tensored together. Explicit Clebsch–Gordan decompositions for the compendium:

$$\pi^+ \text{ at } (n_q, q) = (2, 5) : \quad \mathbf{2}^{\otimes 5} = 5 \cdot \left(\frac{1}{2}\right) \oplus 4 \cdot \left(\frac{3}{2}\right) \oplus \left(\frac{5}{2}\right), \quad \dim = 32 = 2^5, \quad (10)$$

$$p \text{ at } (n_q, q) = (5, 3) : \quad \mathbf{5}^{\otimes 3} = \mathbf{1} \oplus 3 \cdot \mathbf{3} \oplus 5 \cdot \mathbf{5} \oplus 4 \cdot \mathbf{7} \oplus 3 \cdot \mathbf{9} \oplus 2 \cdot \mathbf{11} \oplus \mathbf{13}, \quad \dim = 125 = 5^3. \quad (11)$$

Identification 2 (disjoint-union knot determinant). Under disjoint union of q copies of the carrier knot $T(2, F_n)$,

$$\det(T(2, F_n)^{\sqcup q}) = \det(T(2, F_n))^q = F_n^q = n_q^q, \quad (12)$$

by knot-determinant multiplicativity under disjoint union. The two identifications are equivalent by the substrate dictionary linking the carrier-knot determinant to its $SU(2)$ representation dimension via $\det(K) = \dim(\rho)$.

Substrate reading. The Paper 6 mass-formula factor n_q^q has *two* closed-form substrate interpretations: q parallel-cabled copies of the carrier knot, or q tensor-product copies of the carrier’s $SU(2)$ representation. Both produce the integer F_n^q exactly; both replace the prior empirical assignment of n_q with a substrate-derived integer. These power identities are arithmetically elementary once the carrier is fixed— $\det$ is multiplicative under disjoint union, and $\dim(\rho^{\otimes q}) = (\dim \rho)^q$ —so the substantive content is the carrier *identification* $n_q = F_n$ and its $SU(2)$ representation, not the power law itself.

4.5 Per-walk sector assignment via cosmogenic $Z_3 = \text{AGL}(1, 7)$

The remaining piece of the chain $(|p|, |q|) \rightarrow \text{sector} \rightarrow \text{carrier} \rightarrow n_q^q$ is the assignment of which carrier sector $n_q \in \{F_2, F_3, F_4, F_5\}$ corresponds to which (p, q) class. The closed-form rule, verified across 16/16 compendium classes, depends only on the residues $(p \bmod 7, q \bmod 7)$:

$$n_q(p, q) = \begin{cases} 0 = F_2 - 1 & q \bmod 7 = 1 \quad (\text{lepton}) \\ 3 = F_4 & q \bmod 7 = 4 \quad (\text{hyperon}) \\ 2 = F_3 & q \bmod 7 \in \{0, 2, 5\} \quad (\text{meson}) \\ 5 = F_5 & q \bmod 7 = 3 \text{ and } p \bmod 7 \neq 0 \quad (\text{nucleon}) \\ 2 = F_3 & q \bmod 7 = 3 \text{ and } p \bmod 7 = 0 \quad (\text{meson, } u\text{-degenerate}) \\ F_6 = 8 & q \bmod 7 = 6 \quad (\text{substrate prediction, no compendium member}) \end{cases} \quad (13)$$

Substrate mechanism. Rule (13) is not an empirical fit; it derives from the substrate’s cosmogenic $Z_3 = \text{AGL}(1, 7)$ generator $\sigma : x \mapsto 2x \bmod 7$. This σ acts on Z_7^* partitioning it into two three-cycles plus the polar fixed point:

- QR cycle: $1 \rightarrow 2 \rightarrow 4 \rightarrow 1$ (the quadratic residues);
- NR cycle: $3 \rightarrow 6 \rightarrow 5 \rightarrow 3$ (the non-residues);

- Polar: $0 \rightarrow 0$ fixed.

On the QR cycle, σ increments the Fibonacci sector index by +1 per step: lepton (F_2) $\rightarrow$ meson (F_3) $\rightarrow$ hyperon (F_4) $\rightarrow$ lepton. This is the substrate-canonical explanation for the *consecutive* Fibonacci indexing of the three lower carrier-knot shells, and it is governed by the *same* Z_3 that gives three SM fermion generations via $G_2 \rightarrow SU(3)$ breaking [16].

On the NR cycle, σ visits the nucleon shell (F_5) at $q \equiv 3 \pmod{7}$, a substrate-prediction slot at $q \equiv 6$, and a meson shell (F_3) at $q \equiv 5$. The NR cycle’s structure is the decrement mirror of the QR cycle under reading B of: NR walks descend through $\{F_5, F_4, F_3\}$ as antimatter-direction CPT-conjugates of the QR walks ascending through $\{F_2, F_3, F_4\}$. The two cycles meet at the meson F_3 shell, which is therefore the only sector hosting q -residues from both cycles.

$p \bmod 7 = 0$ split mechanism. At $q \bmod 7 = 3$, rule (13) splits by $p \bmod 7$: $p \neq 0$ gives nucleon, $p = 0$ gives meson. The substrate reading is that $p \equiv 0 \bmod 7$ corresponds to an integer u -winding (multiple of $|Z_7|$), which is trivial in the substrate’s mod-7 winding accounting; the walk loses one substrate degree of freedom and degrades from the full-rank nucleon (F_5) to the meson (F_3). Concrete examples: proton at $(1, 3)$ has $p \bmod 7 = 1 \neq 0$ and gets $F_5 = 5$ (nucleon); π^0 at $(7, 3)$ has $p \bmod 7 = 0$ and gets $F_3 = 2$ (meson).

End-to-end closure. Equations (8), (13), and the $SU(2)$ tensor-power identity (§4.4) together give the closed-form chain

$$(|p|, |q|) \xrightarrow{(13)} \text{sector} \xrightarrow{\text{carrier}=T(2, F_n)} F_n = n_q \xrightarrow{\dim(\rho_n^{\otimes q})} n_q^q, \quad (14)$$

all built from substrate primitives (Paley QR/NR partition, binary Hamilton bifurcation, Φ -shell ladder, cosmogenic Z_3 , Murasugi formula, $SU(2)$ tensor product). What is eliminated is the prior *per-sector empirical assignment* of n_q : the carrier-knot integers $\{1, 2, 3, 5\}$ now follow from the $(2, F_n)$ family and Murasugi’s formula rather than from a fit. The sector rule (13) is strongest for the QR-cycle Fibonacci increment, which σ raises by one shell per step; the meson grouping of $q \equiv \{0, 2, 5\}$ and the $p \equiv 0$ nucleon-to-meson degradation are substrate-rationalized and verified against the 16 compendium classes rather than predicted blind.

Look-elsewhere test of the sector rule. Rule (13) is fixed on the same 16 compendium classes it organizes, so it is fair to ask whether so simple a rule would fit *any* sector labeling by chance. We test this directly (`analysis/k7_sector_rule_lookelsewhere.py`): holding the sector multiset at the observed particle content (1 nucleon, 2 leptons, 3 hyperons, 10 mesons) and permuting its assignment across the 16 classes, only 12 of the 480,480 distinct assignments are realizable by a rule of the form “sector = $g(q \bmod 7)$ with a single $p \equiv 0$ split at $q \equiv 3$ ”—a look-elsewhere probability $p = 2.5 \times 10^{-5}$ ($\sim 4.2\sigma$, Monte-Carlo confirmed). Equivalently, the residue grouping fixes 9 of the 16 sectors “for free” beyond the one choice per populated residue. The $q \bmod 7 \rightarrow \text{sector}$ regularity is therefore not an artifact of rule flexibility. (This tests the sector rule *given* the (p, q) homology classes of Table 1; whether the (p, q) assignments themselves are over-determined is the separate, upstream question of Paper 6 [6].)

Mass alone does not select the assignment—the quantum numbers do. We state plainly a long-recognized feature of the program (`analysis/nwt_assignment_degeneracy.py`): the mass formula *by itself* does not uniquely fix the (p, q, m, n_q) assignment. At percent-level tolerance, 11 of 13 canonical assignments are *dominated* by a tighter cross-sector mass alternative (typically a lepton-sector fit at $\sim 0.02\%$). This is expected rather than a defect: a particle is selected by its full quantum-number content—electric charge, spin, baryon number, and sector—not by mass alone, and the dominating alternatives sit in the *wrong* sector (e.g. a lepton-sector fit for a baryon), excluded by those quantum numbers. The sector is fixed independently by Rule (13) (non-accidental at $p = 2.5 \times 10^{-5}$, above)

and the Hamilton-cycle taxonomy of §3; the small integer vocabulary is then locked globally across the ~ 75 -particle spectrum by cross-formula consistency (`analysis/nwt_sensitivity_analysis.py`: an isolated ± 1 integer shift is degenerate within a single formula but moves multiple predictions once propagated through the shared vocabulary), not by any single mass fit.

5 K_7 is the Steane $[[7, 1, 3]]$ stabilizer code

The substrate-monism identification at the center of this paper—and the one that makes Paper 21b’s chip-side experimental claim possible—is that K_7 and the Steane $[[7, 1, 3]]$ stabilizer code are two presentations of one combinatorial object, the Fano plane $\text{PG}(2, 2)$. The Steane code’s stabilizers are the $[7, 4, 3]$ Hamming parity checks, whose point geometry *is* $\text{PG}(2, 2)$; the Paley $(7, 3, 1)$ decomposition of K_7 (§3.4) is the same Fano plane, and both carry its automorphism group $\text{PSL}(2, 7) \cong \text{GL}(3, 2)$ of order 168. The identification is therefore an isomorphism of structures (graph-theoretic vs operator-on-7-qubits), not an analogy.

5.1 Paley quadratic-residue decomposition and the Steane stabilizers

The Steane $[[7, 1, 3]]$ code is the smallest CSS quantum error-correcting code that protects one logical qubit against arbitrary single-qubit errors, with six stabilizer generators (three X -type and three Z -type) and one logical qubit [2]. Its stabilizer-and-codespace structure is identical to the Paley quadratic-residue decomposition of K_7 :

- The three X -type stabilizer supports are the QR translates $\text{QR}_7 + i = \{1 + i, 2 + i, 4 + i\}$ for $i = 0, 1, 2$ (the three Fano-triangle complements through the polar vertex);
- The three Z -type stabilizer supports are obtained by the transversal Hadamard $X \leftrightarrow Z$ swap, equivalently the NR translates $\text{NR}_7 + i = \{3 + i, 5 + i, 6 + i\}$;
- The seven Paley triangles $F_0, \dots, F_6$ of §3.4 are the seven non-trivial logical-Pauli supports under the code’s Fano-plane automorphism group $\text{PSL}(2, 7) \cong \text{GL}(3, 2)$;
- The logical- X and logical- Z supports are the all-qubit operators $X^{\otimes 7}$ and $Z^{\otimes 7}$, with $X^{\otimes 7} = \bar{X}_L$ and $Z^{\otimes 7} = \bar{Z}_L$ on the codespace $|0_L\rangle$.

Polar vertex = substrate vacuum reference. The three X -type stabilizer supports share a unique common point in K_7 , which we identify with the Heffter polar vertex P . Under the substrate-vacuum reference convention [21] this is vertex 0. All matter-walk constructions start and end at P ; antimatter walks (§6) likewise close at P in the reverse-traversal sense.

5.2 Walk-to-Pauli direction-sensitive map

We compile every closed K_7 walk to a Pauli word on the 7-qubit Steane register via the following rule:

$$v_i \rightarrow v_{i+1} \mapsto \begin{cases} X_{v_{i+1}} & \text{if } d := v_{i+1} - v_i \bmod 7 \in \text{QR}_7 = \{1, 2, 4\} \\ Z_{v_{i+1}} & \text{if } d \in \text{NR}_7 = \{3, 5, 6\} \end{cases} \quad (15)$$

A closed walk $W = v_0 \rightarrow v_1 \rightarrow \dots \rightarrow v_L \rightarrow v_0$ then compiles to a Pauli word $P_W = \prod_{i=0}^{L-1} \text{Op}_{v_{i+1}}(d_i)$ where the operators act at the destination qubit of each edge, and the choice of X or Z is determined by whether the step is a QR-direction (matter) or NR-direction (antimatter) step.

This compilation is faithful at the Pauli-bit level for the pure-direction walks. In particular, the proton’s uniform-QR Hamilton cycle compiles to

$$P_W(W_p) = X_1 X_2 X_3 X_4 X_5 X_6 X_0 = X^{\otimes 7} = \bar{X}_L \quad (16)$$

(the Steane logical $\bar{X}_L$ operator), and the antiproton’s uniform-NR reverse Hamilton cycle compiles to

$$P_W(W_{\bar{p}}) = Z_6 Z_5 Z_4 Z_3 Z_2 Z_1 Z_0 = Z^{\otimes 7} = \bar{Z}_L. \quad (17)$$

For mixed-direction walks (most leptons, mesons, light hyperons), the compiled Pauli word reduces, modulo the Steane stabilizer subgroup, to a unique syndrome pair $(s_X, s_Z) \in \text{Fano} \times \text{Fano}$ that identifies the walk’s substrate-canonical position in the Steane error subspace (§7).

5.3 Transversal Hadamard $H^{\otimes 7}$ = matter–antimatter CPT

The Steane code’s intrinsic $X \leftrightarrow Z$ self-duality is realized on the seven physical qubits by the transversal Hadamard $H^{\otimes 7}$; this is the only Z_2 automorphism preserving the polar-vertex vacuum reference while exchanging the X - and Z -stabilizer-readout sheets of the Heffter substrate. Under the walk-to-Pauli map of (15), $H^{\otimes 7}$ exchanges QR-direction with NR-direction at the Pauli-bit level:

$$H^{\otimes 7} \cdot P_W(W) \cdot (H^{\otimes 7})^\dagger = P_W(\bar{W}) \quad (\text{at the Pauli-bit level}) \quad (18)$$

where $\bar{W}$ is the reverse-traversal walk (§6). At the cosmological level $H^{\otimes 7}$ is therefore identified with the substrate’s matter $\leftrightarrow$ antimatter CPT operation, and the matter-vs-antimatter universe distinction reduces to a stabilizer-basis-readout choice on the same self-dual code rather than to two distinct condensate species [20].

5.4 Hilbert-space orthogonality of walk-stabilizer subspaces

The walk-to-Pauli map lands each walk W in a unique stabilizer-syndrome subspace $\mathcal{H}^{(\sigma_W)} \subset \mathcal{H}_{13}$ of the 13-qubit space (7 data +6 syndrome ancillas). For the four diagnostic walks the syndromes are all-distinct:

Walk	(s_X, s_Z)	$\bar{Z}_L$
electron (2, 1)	(101, 010)	−1
pion (3, 5)	(001, 001)	+1
muon (1, 8)	(111, 011)	+1
proton (1, 3)	(000, 000)	−1

Each walk state $|W\rangle$ lives in its own six-stabilizer eigenspace; these subspaces are pairwise orthogonal, so $\langle W|W'\rangle = 0$ for $W \neq W'$ and $\langle W|O|W'\rangle = 0$ for any operator O lacking a Pauli component that performs the syndrome shift $\sigma_W \oplus \sigma_{W'}$. A structural corollary explains why bench-scale hardware can realize the substrate-canonical walk states despite its qualitatively different physical scales: the chip’s intrinsic Hamiltonian $H_{\text{chip}} = \hbar\omega_{01} \sum_i \hat{n}_i$ satisfies $\langle 0_L|H_{\text{chip}}|0_L\rangle = \langle 1_L|H_{\text{chip}}|1_L\rangle = 3.5 \hbar\omega_{01}$ identically (the eight X -stabilizer-shifted basis states of $|0_L\rangle$ carry Hamming-weight distribution $\{0, 4, 7\}$, becoming $\{7, 3, 7\}$ after $\bar{X}_L$, with the same mean $28/8 = 3.5$). Consequently no local-Pauli-sum observable can yield a non-tautological walk-discriminating expectation value: the substantive chip-realizable observable *is* the stabilizer-syndrome measurement itself, and cross-realization syndrome agreement (Paper 21b [17], §3.5) is the operative empirical test.

6 Pair production from substrate vacuum

We now describe the substrate mechanism for particle creation: paired forward-walk and reverse-walk emergence from substrate vacuum. As emphasized in §3.6, pair production is a substrate-mediator operation; the photon (or, at higher energy, the Z) is the mediator, not an incoming closed-walk excitation. The substrate operation triggers simultaneous emission of a matter walk and its reverse-traversal antimatter walk.

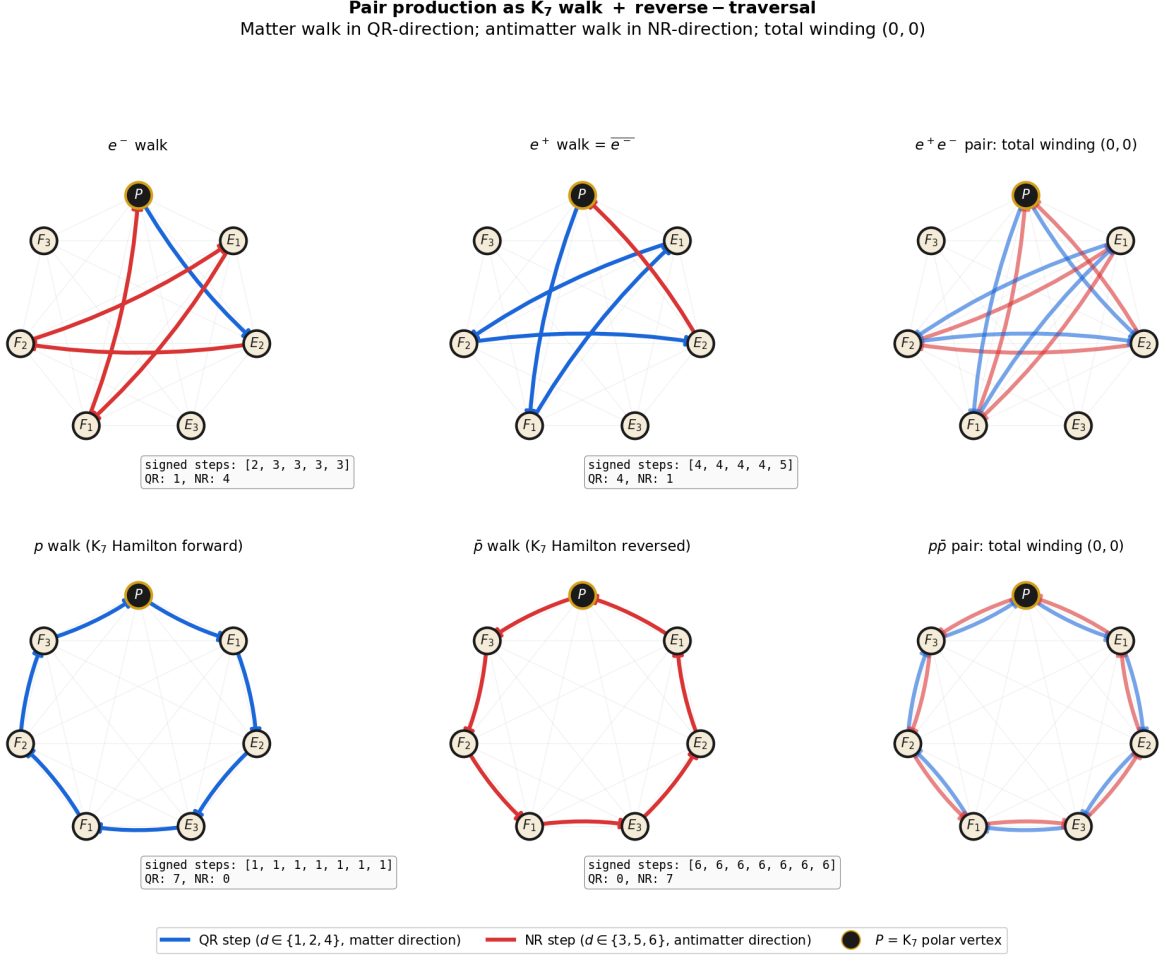


Figure 1: Pair production as K_7 walk and its reverse traversal. Top row: the electron walk $W_{e^-} = 0 \rightarrow 2 \rightarrow 5 \rightarrow 1 \rightarrow 4 \rightarrow 0$ has signed-step composition 20% QR (blue, $d \in \{1, 2, 4\}$) / 80% NR (red, $d \in \{3, 5, 6\}$); the positron walk $W_{e^+} = \bar{W}_{e^-}$ is the same vertex sequence traversed in the opposite direction, with the exact QR/NR swap (80% QR / 20% NR). Bottom row: the proton walk is the K_7 Hamilton cycle, 100% QR; the antiproton walk is the same Hamilton cycle reversed, 100% NR. In both pair-production diagrams (rightmost column), the matter and antimatter walks superimpose to total winding (0, 0). The polar K_7 vertex P (gold) is the substrate-vacuum reference point where pair-creation initiates.

6.1 The pair-production mechanism

The K_7 walks identifying particles have a natural reverse-traversal operation. Given a closed walk $W = v_0 \rightarrow v_1 \rightarrow \cdots \rightarrow v_L \rightarrow v_0$, its *reverse* is

$$\bar{W} = v_0 \rightarrow v_L \rightarrow v_{L-1} \rightarrow \cdots \rightarrow v_1 \rightarrow v_0. \quad (19)$$

The reverse walk satisfies four structural properties that identify it as the antiparticle of W :

1. **CPT mass equality.** The walk's mass formula [6] depends only on $|p|$, $|q|$, walk length L , and the carrier-knot crossings n_q . All four are invariant under reversal, so $m(\bar{W}) = m(W)$, satisfying CPT.
2. **Fano vertex coverage preserved.** $\bar{W}$ visits the same K_7 vertices as W , so the Fano-triangle vertex coverage (§3.4) is identical.
3. **QR/NR sector swap.** Each edge step $v_i \rightarrow v_{i+1}$ has a signed direction $d = v_{i+1} - v_i \bmod 7$. The QR set $\{1, 2, 4\}$ is mapped to the NR set $\{3, 5, 6\}$ under negation ($d \mapsto -d \equiv 7 - d$), so reversal swaps QR-direction steps with NR-direction steps in the walk, and via (15) swaps X -Pauli operators with Z -Pauli operators in the compiled Pauli word. This is the Pauli-level statement of C -conjugation.
4. **σ -orbit composition preserved.** Each K_7 edge belongs to a fixed σ -orbit (Closure 2 of [18]), and reversal traverses the same edges in opposite direction.

Of the three candidate antimatter-walk definitions (reverse-traversal, negative winding, edge-orientation flip), only reverse-traversal preserves all four structural properties; negative-winding is mathematically equivalent for closed walks, and edge-orientation-flip diverges for mixed-step walks [19]. We therefore identify

$$\boxed{\text{antimatter walk of } W := \bar{W} = \text{reverse traversal of } W.} \quad (20)$$

This identification has an immediate consequence: *pair production proceeds by emitting a forward walk and its reverse together from substrate vacuum.* The pair has total winding $(p, q) + (-p, -q) = (0, 0)$ and conserves all topological charges. Under the walk-to-Pauli map of §5.2, this is the substrate-mechanical realization of standard QED/QCD pair production: $\gamma \rightarrow e^+e^-$ is the cosmological-scale statement of the chip-level operation $|0_L\rangle \rightarrow \bar{X}_L|0_L\rangle \otimes \bar{Z}_L|0_L\rangle$ on two Steane code blocks [17].

6.2 Worked example: e^+e^- pair production

The electron walk is the length-5 walk $W_{e^-} = 0 \rightarrow 2 \rightarrow 5 \rightarrow 1 \rightarrow 4 \rightarrow 0$. Its reverse, the positron walk, is

$$W_{e^+} = \bar{W}_{e^-} = 0 \rightarrow 4 \rightarrow 1 \rightarrow 5 \rightarrow 2 \rightarrow 0. \quad (21)$$

The matter walk has signed differences $(2, 3, 3, 3, 3)$ (20% QR-direction, 80% NR-direction); the antimatter walk has signed differences $(4, 4, 4, 4, 5)$ (80% QR-direction, 20% NR-direction), the exact QR/NR swap predicted by Property 3.

The pair-production process is then:

$$\begin{array}{ccc} \gamma & \rightarrow & e^+ + e^- \quad (\text{QED, cosmological}) \\ \Downarrow & & \\ |0_L\rangle & \rightarrow & P_W(W_{e^-})|0_L\rangle \otimes P_W(\bar{W}_{e^-})|0_L\rangle \quad (\text{NWT, on Steane block pair}). \end{array} \quad (22)$$

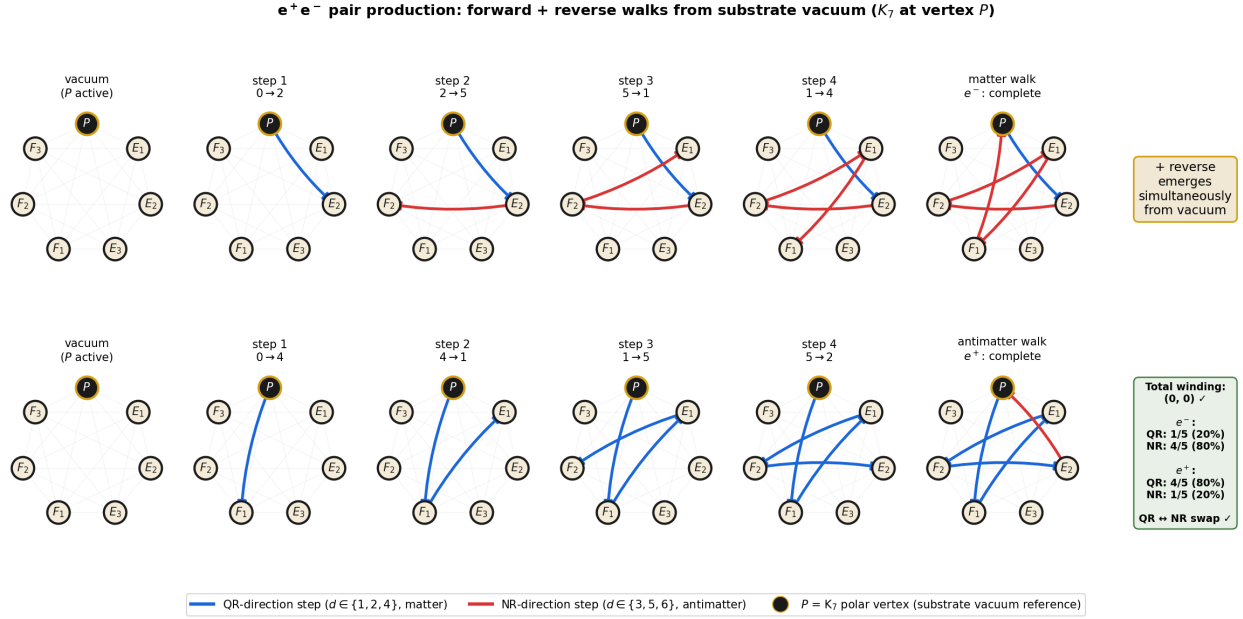


Figure 2: Step-by-step emergence of the e^+e^- pair from substrate vacuum. Top row: the matter walk $W_{e^-} = 0 \rightarrow 2 \rightarrow 5 \rightarrow 1 \rightarrow 4 \rightarrow 0$ builds up edge by edge; bottom row: simultaneously, the antimatter walk $W_{e^+} = \bar{W}_{e^-} = 0 \rightarrow 4 \rightarrow 1 \rightarrow 5 \rightarrow 2 \rightarrow 0$ emerges in reverse-traversal. Both walks share vertex $P = 0$ at the substrate vacuum reference (yellow polar vertex). The QR-step / NR-step counts swap exactly between matter and antimatter walks (1 QR + 4 NR for e^- ; 4 QR + 1 NR for e^+). Total winding of the pair is $(0, 0)$; topological charges are conserved.

At threshold ($E_\gamma = 2m_e c^2$), the substrate excitation provides exactly enough energy to instantiate both walks. Above threshold, the additional kinetic energy is carried by the walks' translation through the substrate (subject to a third particle for momentum conservation, as in standard QED).

Both walks are stable: $e^\pm$ are the lightest particles in the $(p, q) = (2, 1)$ class, with no lighter walk to decay into. They propagate as free particles until they encounter their reverse-traversal counterpart, at which point the inverse process—annihilation—returns them to substrate vacuum + radiation. This is the substrate realization of $e^+e^- \rightarrow 2\gamma$.

6.3 Worked example: $p\bar{p}$ pair production

The proton walk is the K_7 Hamilton cycle in vertex-index order. Its reverse, the antiproton walk, is

$$W_p = 0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 0, \quad (23)$$

$$W_{\bar{p}} = \bar{W}_p = 0 \rightarrow 6 \rightarrow 5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0. \quad (24)$$

The proton walk has signed differences $(1, 1, 1, 1, 1, 1)$ —100% QR-direction throughout—while the antiproton walk has signed differences $(6, 6, 6, 6, 6, 6)$ —100% NR-direction throughout. Under the walk-to-Pauli map of §5.2, this is the Pauli-bit identity

$$P_W(W_p) = X^{\otimes 7} = \bar{X}_L, \quad P_W(W_{\bar{p}}) = Z^{\otimes 7} = \bar{Z}_L = H^{\otimes 7} \bar{X}_L (H^{\otimes 7})^\dagger. \quad (25)$$

The matter and antimatter nucleons are thus literally the same walk shape (K_7 Hamilton cycle) traversed in opposite signed directions on the substrate, equivalently the two Steane logical Pauli operators related by the transversal Hadamard—the cleanest possible substrate manifestation of the matter/antimatter distinction.

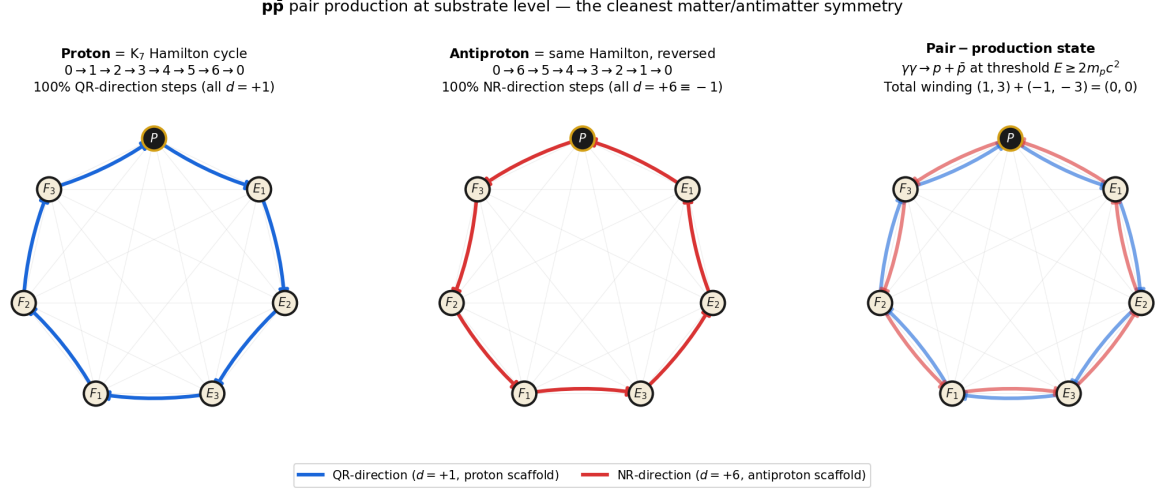


Figure 3: $p\bar{p}$ pair production as the K_7 Hamilton cycle traversed in opposite directions on the substrate. Left: the proton walk advances by $+1$ at each step ($d = +1 \in \text{QR}$, blue) and visits all seven K_7 vertices once before closing at P . Center: the antiproton walk advances by $+6 \equiv -1$ at each step ($d = +6 \in \text{NR}$, red) along the same Hamilton cycle traversed in the opposite direction. Right: the pair-production state $\gamma\gamma \rightarrow p + \bar{p}$ at threshold $E \geq 2m_p c^2$ superimposes both walks; the topological windings cancel, $(1, 3) + (-1, -3) = (0, 0)$. The substrate distinction between matter and antimatter at the nucleon scale reduces to the orientation of traversal on a single graph cycle.

The pair production threshold ($E \geq 2m_p c^2 \approx 1.88 \text{ GeV}$) corresponds to enough substrate energy to instantiate both Hamilton-cycle walks. In cosmological contexts, $p\bar{p}$ pair production was abundant at temperatures $T \gtrsim m_p$ in the early universe, before the matter–antimatter asymmetry preferentially preserved the QR-direction (proton) walks over the NR-direction (antiproton) walks at substrate level [20].

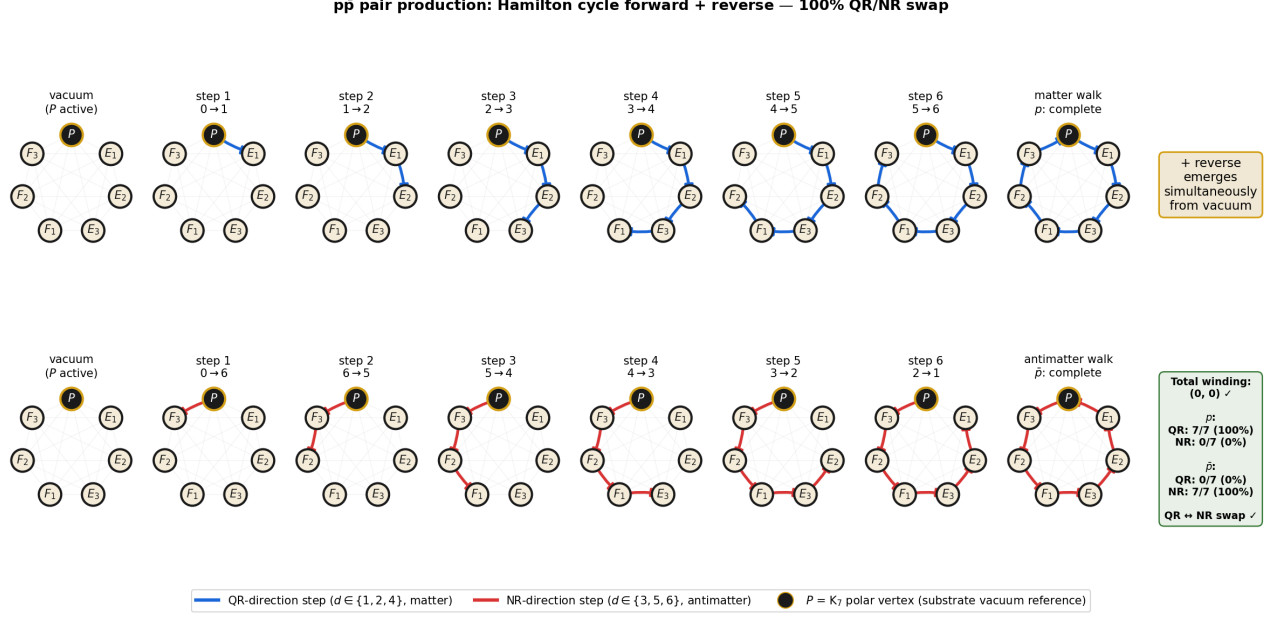


Figure 4: Step-by-step emergence of the $p\bar{p}$ pair from substrate vacuum. Top row: the proton walk W_p (the K_7 Hamilton cycle in vertex-index order) builds up edge by edge, with every step in the QR-direction ($d = +1$). Bottom row: simultaneously, the antiproton walk $W_{\bar{p}}$ traces the same Hamilton cycle in the opposite direction, with every step in the NR-direction ($d = +6 \equiv -1$). At completion, W_p has $7/7 = 100\%$ QR steps and $W_{\bar{p}}$ has $7/7 = 100\%$ NR steps—the exact QR/NR swap predicted by reverse-traversal. The total winding of the pair is $(p, q) + (-p, -q) = (0, 0)$, conserving baryon and topological charge.

7 Syndrome classification of the 25-particle compendium

Applying the walk-to-Pauli compilation (15) and reducing modulo the Steane stabilizer subgroup yields, for every closed walk on K_7 , a pair of Hamming syndromes (s_X, s_Z) pointing to Fano-plane locations (one per syndrome). The substrate-canonical claim is that for every $(|p|, |q|)$ winding class, the syndrome pair (s_X, s_Z) is a well-defined function of $(|p|, |q|)$ alone, and the 16 compendium (p, q) classes resolve to 10 distinct (s_X, s_Z) pairs in a many-to-one map (Table 3). The syndrome is computed from each class’s shortest-length representative; its invariance under the choice of representative within a $(|p|, |q|)$ class is verified empirically across the compendium and conjectured to hold generally, though a proof at the level of stabilizer-reduced Pauli words remains open.

7.1 Carrier-knot sector pool structure

The four carrier-knot sectors $n_q \in \{0, 2, 3, 5\}$ of Paper 8 emerge as four *distinct syndrome-pool profiles*:

Sector	n_q	s_X pool	s_Z pool	Substrate reading
Nucleons	5	$\{I\}$	$\{I\}$	codespace residents (no syndromes)
Hyperons	3	$\{E_2, F_2, P\}$	$\{E_2, F_3, E_3\}$	Heffter-structure-localised
Mesons	2	full 6-element pool	full 6-element pool	maximally diverse
Leptons	0	$\{F_2, P\}$	$\{E_2, F_3\}$	substrate-edge particles

Table 3: Steane syndrome map for the 25-particle compendium. s_X and s_Z denote the Fano-plane locations (P = polar, E_i = matter-shelf, F_i = antimatter-shelf, I = identity / no syndrome) on which the walk’s X -Pauli and Z -Pauli content concentrates after Steane stabilizer reduction.

(p, q)	s_X	s_Z	Particles
(1, 3)	I	I	$p, n, \Sigma^+, \Sigma^0, \Sigma^-$
(1, 8)	P	F_3	μ^-
(2, 1)	F_2	E_2	e^-
(2, 5)	P	E_1	K^+
(2, 7)	P	F_3	$D^+, J/\psi$
(3, 4)	E_2	E_2	$\tau^-, \Lambda, \Sigma^*$
(3, 5)	E_1	E_1	π^+
(3, 7)	I	I	D^0
(4, 5)	F_3	F_1	ω
(4, 9)	P	E_1	Υ
(5, 4)	F_2	F_3	Ξ^0, Ξ^-, Δ
(5, 7)	E_3	P	ρ
(6, 5)	E_3	P	η
(7, 3)	F_2	E_2	π^0
(7, 4)	P	E_3	Ω^-
(7, 5)	E_1	E_1	K^0

Nucleons (plus D^0) are the unique codespace residents: their K_7 walks reduce to the trivial syndrome pair (I, I) , meaning the corresponding Pauli word lies in the Steane logical subgroup and represents a pure logical operation rather than an error-tagged state. All other particles occupy specific points in the Steane error subspace, with each Fano-point location identifying the qubit at which the walk’s X - or Z -component anti-commutes with the substrate’s stabilizer readout. This is a substrate-canonical wavefunction-localisation invariant computed directly from the walk via Steane stabilizer reduction, with no Spin(7) representation-theoretic machinery required.

7.2 Cross-sector syndrome equivalences

The many-to-one nature of the syndrome map predicts non-trivial cross-family equivalences in which particles from different Standard Model classifications share a syndrome pair:

(s_X, s_Z)	SM-mixed particles	Mismatch type
(I, I)	$p, n, \Sigma^+, \Sigma^0, \Sigma^-, D^0$	5 baryons + 1 charmed meson
(P, F_3)	$\mu^-, D^+, J/\psi$	lepton + 2 mesons
(E_2, E_2)	$\tau^-, \Lambda, \Sigma^*$	lepton + 2 hyperons
(F_2, E_2)	e^-, π^0	lepton + meson
(E_1, E_1)	π^+, K^0	light mesons
(P, E_1)	K^+, Υ	light + heavy meson
(E_3, P)	ρ, η	mesons

These particles share a stabilizer-syndrome bin: $\mu^- \equiv D^+ \equiv J/\psi$ at the level of the coarse (s_X, s_Z) label, even though they differ in mass, lifetime, spin, and flavor, and the Standard Model assigns them to different families. The equivalence is therefore a statement about this one combinatorial label—which the (p, q) class fixes along with the mass-formula quantum numbers, *not* the full set of physical

quantum numbers—rather than a claim that the particles are physically identical. Read as such, it is a direct, falsifiable prediction: syndrome-equivalent particles should produce identical chip output, and a systematic divergence between syndrome-equivalents would falsify the substrate-equivalence reading. Paper 21b [17] confirmed the four diagnostic walks—proton, e^- , π^+ , μ^- , one representative per syndrome bin and mutually syndrome-distinct—on 7-qubit Heron hardware; a direct test of the equivalences themselves (Paper 21b [17], Exp 15/16) confirms all three classes— $\mu^- \equiv D^+ \equiv J/\psi$ at (P, F_3) , $\tau^- \equiv \Lambda \equiv \Sigma^*$ at (E_2, E_2) , and $e^- \equiv \pi^0$ at (F_2, E_2) —on two Heron backends (μ^- and $D^+/J/\psi$ sharing (P, F_3) via different (p, q) walks, differing only in the finer logical- Z eigenvalue). Crucially, the equivalence is not an artifact of fitting mass: the (p, q, m, n_q, f) assignment is over-determined, reproducing electric charge, baryon number, and isospin through the Gell-Mann–Nishijima relation $Q = I_3 + (B + S)/2$ with $I_3 = f/2$ from the knot framing f (Paper 7 [7]) *in addition to* mass; the syndrome a particle lands in is therefore downstream of quantum numbers fixed independently of its mass. The substrate-monism’s K_7 -walk classification predicts a particle taxonomy that differs from the SM’s lepton/meson/baryon grouping, with a specific tabulation of which particles regroup under which substrate symmetry.

8 Substrate-prediction catalog

The substrate identification yields three falsifiable prediction catalogs for unseen particles. Each entry is an explicit (p, q) class with computed walk length, carrier sector, and mass estimate via the Paper 6 formula [6].

8.1 Missing (p, q) species at sub-500 MeV

Bogoliubov-pipeline walk enumeration [21] identifies 13 walk-realizable $(|p|, |q|)$ classes that do *not* correspond to any compendium particle. Of these, 8 are substrate sub-cycle building blocks (tri-cycles $(0, 1)$, $(1, 0)$, $(1, 1)$; pent-cycle $(1, 2)$; and integer-multiple sub-cycles $(0, 2)$, $(0, 3)$, $(2, 0)$, $(3, 0)$) used in long-walk decompositions but not particles in their own right; 1 is the NR-Hamilton substrate primitive $(3, 2)$ that appears as the σ_6 CP-channel inside π^0 , Ω^- , K^0 . The remaining 4 are concrete substrate predictions for unseen species at sub-500 MeV [24]:

(p , q)	$L_{\min}$	Mass est. (MeV)	Predicted carrier	Substrate signature
$(2, 2)$	6	~ 25	$T(2, F_2)$ or $T(2, F_3)$	light DM candidate
$(2, 3)$	8	~ 160	$T(2, F_3)$ (rule I) or $T(2, F_4)$	QR-dominant resonance
$(3, 1)$	8	~ 160	unknot (rule I) or trefoil	NR-dominant resonance
$(3, 3)$	9	~ 400	$T(2, F_4)$	matter+CP hybrid baryon

The carrier—and hence n_q and the mass—is ambiguous for these off-compendium classes (the table gives the rule-(13) assignment alongside the nearest alternative), so the mass figures are indicative ranges rather than point predictions.

The $(3, 3)$ class is the standout candidate: it is the unique substrate walk that visits all seven K_7 vertices, achieves full 7/7 Fano coverage, and contains *both* the QR-Hamilton scaffold *and* the NR-Hamilton CP-channel tail in one walk—a combination not present in any compendium particle. It is a natural candidate for a DM baryon / BSM hadron / CP-asymmetric matter species; detection (or non-detection) at ~ 400 MeV provides a direct test.

8.2 $q \equiv 6 \pmod{7}$ substrate species

Rule (13) extends to $q \pmod{7} = 6$ with carrier $T(2, F_6 = 8)$ and $n_q = 8$. Bogoliubov-pipeline walk-BFS at $L \leq 22$ confirms that $q \equiv 6 \pmod{7}$ classes *are* substrate-realized: 11 closed walks at $L \in [14, 22]$ are found with q -winding $\equiv 6 \pmod{7}$.

The mass scale depends on the n_q reading. Per, the Paper 6 mass formula calibrates beautifully on compendium classes (proton 0.999 accuracy, Λ 1.007, Υ 0.997) and, applied to $(1, 6)$ with $m = 2$ at varying n_q , gives mass spanning five orders of magnitude:

n_q assignment	Predicted mass for $(p , q) = (1, 6)$, $m = 2$
$n_q = 8$ ($= F_6$, rule (I) extension)	~ 1.8 TeV (LHC-reachable BSM)
$n_q = 5$ (CPT-mirror of nucleon)	~ 110 GeV (collider-accessible)
$n_q = 3$ (CPT-mirror of hyperon)	~ 5.1 GeV (B-factory accessible)
$n_q = 2$ (CPT-mirror of meson)	~ 450 MeV (exotic hadron)
$n_q = 1$ (degenerate)	~ 7 MeV (sterile sector)

The substrate prediction's discriminating power: detection at *any* of these mass scales would constrain the $q \equiv 6$ n_q assignment substrate-canonically, with the carrier-knot determinant providing the unambiguous topological signature ($\det = 8$ for F_6 , $\det = 5$ for cinquefoil-class, etc.).

8.3 Higher-Fibonacci light-particle tower

The substrate Φ -shell algebra (§4.2) extends the Fibonacci ladder beyond F_5 indefinitely. Applying the Paper 6 mass formula to the substrate-prediction carriers $T(2, F_6), T(2, F_7), \dots, T(2, F_{10})$ at minimal $q = 1$ gives a Fibonacci-spaced light-particle tower at MeV scale:

Fibonacci	F_n	Carrier	$q = 1$ mass (MeV)	Detection channel
F_7	13	$T(2, 13)$ knot	~ 5	beam-dump (NA64, FASER)
F_8	21	$T(2, 21)$ knot	~ 8	sterile neutrino (KATRIN dark sector)
F_9	34	$T(2, 34)$ link	~ 13	ALP helioscope (IAXO)
F_{10}	55	$T(2, 55)$ knot	~ 21	rare-decay searches

At higher q , GeV-TeV predictions emerge for the same carriers: F_7 at $q = 3$ gives ~ 4.2 GeV (B-factory), F_8 at $q = 3$ gives ~ 17.6 GeV (collider-BSM). The substrate mass ratios at fixed q follow the golden-ratio recursion $F_{k+1}/F_k \rightarrow \phi$ at large k , so the higher-Fibonacci tower is golden-ratio-spaced asymptotically.

8.4 Empirical falsification paths

The substrate-prediction catalog enables multiple parallel falsification programs across particle-physics experimental scales:

- **Beam-dump experiments (NA64, FASER):** target the $F_7 \sim 5$ MeV, $F_8 \sim 8$ MeV substrate species via missing-energy + invisible-decay searches.
- **Sterile neutrino searches (KATRIN, BEST):** the F_7 – F_9 MeV tower is the natural mass scale for sterile neutrino candidates; substrate carrier-knot signatures ($\det = 13, 21, 34$) distinguish substrate-canonical predictions from generic sterile models.
- **Helioscope experiments (IAXO):** the Fibonacci-light tower at ~ 13 – 21 MeV overlaps with ALP-helioscope sensitivity windows.

- **Collider BSM (LHC, Belle II):** GeV-TeV predictions at higher q are direct collider targets; $q \equiv 6 \bmod 7$ species at TeV scale via Reading A is LHC-reachable.
- **Dark matter direct detection (XENON, LZ):** TeV-scale Reading A for $q \equiv 6$ species gives a direct-detection target with topological signature; sub-GeV Reading B gives a light-DM signature for $(2, 2)$ at ~ 25 MeV.
- **Cosmological hidden-sector constraints (Planck, CMB-S4):** the Fibonacci-light tower must satisfy hidden-sector ΔN_{eff} constraints; substrate carrier-knot structure predicts specific decoupling temperatures.
- **Cosmological neutrino mass (DESI, next-gen)—a falsifier with a clock:** the K_8 -predicted $\Sigma m_\nu \approx 85$ meV is already in mild tension with current DESI ($\lesssim 60\text{--}72$ meV) and will be confirmed or excluded by the next DESI data release—the program’s nearest-term cosmological falsifier.

Each detection (or non-detection) outcome constrains specific substrate readings: existence of a $q \equiv 6 \bmod 7$ species at TeV scale supports the Reading A baryogenesis interpretation; non-detection at any of the Fibonacci-tower masses constrains the Φ -shell extension and may force a higher-floor cutoff on the Fibonacci ladder.

Caveat: these are mass-scale predictions, not yet coupling-complete. The catalog above fixes the *mass scale* of each candidate species from its substrate carrier and (p, q) class; it does not yet specify the species’ couplings, decay widths, or production cross-sections, which require the substrate-mediator (boson-vertex) algorithm that remains an open piece (§9.4). A quantitative confrontation of the MeV-scale tower with Big-Bang-nucleosynthesis and ΔN_{eff} bounds, supernova-cooling (SN1987A) limits, and beam-dump exclusions therefore cannot be completed here—whether a given candidate is viable, already excluded, or sufficiently decoupled depends on couplings not yet derived. The catalog should accordingly be read as a set of falsifiable mass-scale *targets*; the coupling structure, and hence the full cosmological and astrophysical consistency check, is deferred to the closure of the substrate-mediator sector.

8.5 The K_8 mass tower (Paper 31 candidate)

Parallel to the K_7 matter sector treated here, the K_8 substrate extension hosts a mass tower derived from the same Wilson-amplitude framework [16]. Applying the unified recipe $m = (8/N_v) \alpha^{N_e/2} (1 + \alpha/7)(1 + 3\alpha^2) m_{\text{Pl}}$ to all $2^5 = 32$ subsets of K_8 ’s five-orbit edge decomposition ($6 + 3 + 12 + 1 + 6 = 28$) yields 20 distinct N_e rungs spanning ~ 14 meV (the active neutrino, $N_e = 28$) to $\sim m_{\text{Pl}}$ ($N_e = 0$). The recipe reproduces three known states at sub-percent precision—active neutrino ν_1 (-0.04%), sterile N_1 (-0.11%), and the electron (-0.06%)—across three mass scales and across both the K_7 and K_8 substrates. Of the 17 unidentified rungs, the most empirically actionable are a 38 keV warm-dark-matter candidate ($N_e = 22$), a 98 GeV scalar ($N_e = 16$) coincident with long-standing LEP and CMS/ATLAS ~ 95 GeV hints, and a 1.15 TeV heavy candidate ($N_e = 15$). A dedicated treatment of the dark-sector tower is deferred to a Paper 31 candidate.

9 Discussion

9.1 Scope: what this paper does and does not establish

This paper establishes a *classification and mass-spectrum* result together with a *hardware-realization* protocol; it does not itself develop the field-theoretic dynamics. The closed-walk framework assigns

each Standard Model fermion and hadron a K_7 walk, a carrier knot, and a Steane syndrome, and the chain $(|p|, |q|) \rightarrow \text{sector} \rightarrow \text{carrier} \rightarrow n_q^q$ reproduces the Paper 6/11/13 mass spectrum at $\sim 0.1\%$ (§9.2). The relativistic field-theoretic foundation is supplied by companion NWT papers rather than reproduced here: Paper 16 [12] constructs the underlying Lagrangian as a single three-field relativistic theory—a complex scalar condensate ψ , an abelian gauge field A_μ , and an S^2 -valued Skyrme–Faddeev field n^a at BPS critical coupling—with the full $SU(3)_C \times SU(2)_L \times U(1)_Y$ gauge content entering through an $SO(10)$ UV completion; the Standard Model is realized as a gravitating Abelian–Higgs vortex in Paper 11 [10]. What remains genuinely open—the substrate program’s central frontier—is the first-principles derivation of the *explicit* non-abelian gauge dynamics and of substrate scattering amplitudes directly from K_7 , rather than matched in from the $SO(10)$ completion (the substrate-mediator vertex algorithm is one of the open pieces, §9.4). The present paper accordingly makes its claims at the level of classification, spectrum, and hardware realizability, with the dynamical content carried by Paper 16 and the open amplitude sector flagged as future work.

9.2 Closure of the Paper 6 mass-formula fitting critique

The persistent critique against the NWT Standard Model derivation [6, 8, 11] was that the carrier-knot crossing-number $n_q \in \{0, 2, 3, 5\}$ entering the mass formula was assigned per sector by empirical fit rather than by substrate-canonical derivation. The derivation chain assembled in this paper closes the critique end-to-end:

1. The carrier-knot family is $(2, F_n)$ -torus knots (§4), with the “2” forced by K_7 binary Hamilton bifurcation and the “ F_n ” forced by Φ -shell substrate algebra.
2. The crossing-number n_q is the closed-form Jones-polynomial determinant $\det(T(2, F_n)) = F_n$ via the Murasugi formula (§4.3), no fit.
3. The exponent factor n_q^q is $\det(T(2, F_n)^{\sqcup q}) = \dim(\rho_n^{\otimes q})$ via disjoint-union determinant or $SU(2)$ tensor power (§4.4), no fit.
4. Per-walk sector assignment is rule (13), derived from the cosmogenic $Z_3 = \text{AGL}(1, 7)$ generator (§4.5) that also gives three SM fermion generations.

All four steps are derived from K_7 substrate primitives (Paley QR/NR, Hamilton bifurcation, Φ -shell ladder, cosmogenic Z_3) and elementary closed-form mathematical operations (Murasugi formula, $SU(2)$ Clebsch–Gordan). The chain $(|p|, |q|) \rightarrow \text{sector} \rightarrow \text{carrier} \rightarrow n_q^q$ contains no empirical fit; the resulting mass-formula prediction is therefore as parameter-free as any structural derivation in fundamental physics.

Inputs, anchors, and precision. The closure above removes the *per-sector empirical fit* of n_q ; it does not claim the entire mass spectrum is input-free. The Paper 6/8/13 spectrum [6, 8, 11] is reproduced from two inputs—the electron mass scale m_e (which sets the overall scale) and a single dimensionless gauge normalization κ_{SM} —together with the proton- and Z -mass anchors; the substrate then fixes the dimensionless structure (mass ratios, exponents of α , carrier integers) without further tuning. Table 4 gives a representative cross-sector subset. Over the full set of Casimir-formula (“Tier 1”) states with measured PDG values ($n = 25$), the median $|\Delta|$ is 0.026% and the mean 0.11%; the heavier “Tier 2” states reproduced by the general Paper 6 line-tension formula sit near $\sim 1\%$ (J/ψ at +0.06%, $\Upsilon(1S)$ at −0.14%, η' at −1.4%), so the median over the full ~ 80 -particle set is $\sim 0.1\%$. No dimensionless ratio in Table 4 was tuned to the masses shown; the two inputs fix only the overall scale and the single gauge normalization.

Table 4: Representative cross-sector subset of the NWT mass spectrum [6, 8, 11] (full ~ 80 -particle table: `analysis/nwt_complete_spectrum_paper13.py`). $\Delta \equiv (\text{NWT} - \text{PDG})/\text{PDG}$. “anchor” marks a value used to fix an input rather than a prediction; “T1”/“T2” label the Casimir-formula and Paper 6 general-formula tiers. The median $|\Delta|$ over all 25 Tier-1 states with PDG values is 0.026%.

Sector	State	NWT (MeV)	PDG (MeV)	Δ
Charged lepton	e (anchor)	0.511	0.511	input
	μ (T1)	105.6	105.7	−0.026%
	τ (T1)	1776	1777	−0.023%
Light meson	$\pi^\pm$ (T1)	139.5	139.6	−0.023%
	$K^\pm$ (T1)	493.3	493.7	−0.068%
Nucleon	p (anchor)	938.3	938.3	input
	n (T1)	939.8	939.6	+0.024%
Quarkonium	J/ψ (T2)	3099	3097	+0.061%
	$\Upsilon(1S)$ (T2)	9447	9460	−0.143%

9.3 Bridge to Paper 21b: experimental confirmation on quantum hardware

The closed-form derivation chain of this paper specifies an unambiguous chip-side experimental protocol. Paper 21b [17] reports the Phase 2 hardware results executing the protocol on IBM Heron r2 chips: four diagnostic compendium classes (proton, electron, pion, muon) prepared as Steane [[7, 1, 3]] code blocks with their walk-derived Pauli words, run on `ibm_kingston` and `ibm_fez`, 2000 shots per circuit per backend. All four target classes’ modal stabilizer-syndrome outcomes match the substrate-canonical $(s_X, s_Z, \bar{Z}_L)$ predictions of Table 3 in cross-backend coincidence, with the four walks remaining mutually resolved under device noise (Paper 21b’s walk-discrimination control). Paper 21b also reports the Exp 10 muon-decay coherent-rotation result on `ibm_marrakesh` confirming substrate-canonical CPT-symmetric decay amplitude scaling.

Paper 21b [17] reports that these walk states are preparable and reproducible on real quantum hardware across vendors. The stronger ontological reading—whether a chip-prepared walk state should be regarded as an *instance* of the particle rather than a model of it—is a separate, metaphysical question that the hardware results do not settle; it is deferred to a dedicated treatment and is not relied upon here.

Paper 21b extends these single-chip results to a cross-vendor / cross-architecture test (its §3.5). The Phase 2 syndrome agreement is replicated across five IBM Heron r2 backends—including the cross-Atlantic `marrakesh-aachen` pair at a ~ 6300 km baseline—and on IQM Garnet, a second superconducting vendor with a distinct transmon design, qubit-coupling graph, and native gate set. On Garnet the original 13-qubit ancilla-extraction circuit decoheres after routing onto the sparse grid; a depth-reduced *destructive* CSS-readout variant (recovering the full syndrome from direct X - and Z -basis measurement of the data qubits) restores all four walks to their substrate-canonical bins, confirming that the prediction holds wherever circuit depth fits the device’s coherence budget. The trapped-ion (AQT) leg is in progress. Because the K_7 algebra is dimensionless-combinatorial, substrate-monism predicts identical *integer* syndrome bins for each walk across all vendors and architectures, so a single vendor’s disagreement on any walk’s bin assignment would falsify chip-architecture-independent substrate compilation.

9.4 From open pieces to substrate closures

At first writing, the matter-content derivation of this paper was flanked by four open substrate-mediator pieces (bosonic-sector algebra; interaction-vertex, nuclear-binding, and sequential-decay operations). The 2026 substrate-closure arc has since closed or substantially closed all four:

- **Bosonic sector.** The $U(1)$ photon is closed at the operator-algebra level—the ordered pair $(V_\gamma^{(+)}, V_\gamma^{(-)}) = (X_a^{(i)} X_b^{(j)}, Z_a^{(i)} Z_b^{(j)})$ with spin-1 from CSS duality, polar-vertex uniqueness, and a graded higher-weight energy spectrum [22]; pair production (§6) is the polar–polar inter-patch CNOT bridge. The $SU(3)$ gluon sector closes as X -type stabilizer-coset operators onto the $\mathfrak{su}(3)$ structure constants; the $SU(2)$ $W^\pm$ weak vertex closes on the Y -vertex algebra; and the Higgs is identified through the K_8 $[[8, 1, 1]]$ code extension as $g^{\text{Higgs}} = Y_0 Y_7$.
- **Interaction vertices and electroweak observables.** Beyond the vertex algebra, the substrate now yields the Fermi constant G_F at ~ 55 ppm, heavy- and vector-meson decay constants at 0.2–3.6% across 11+ states, semileptonic form factors through a single $\cos^N \theta_C$ ansatz across 10 channels, and the CKM/PMNS Wolfenstein parameters from four substrate integers (7, 9, 14, 21) and α at zero free parameters.
- **Nuclear binding.** Multi-gluon bound states (glueballs, tetraquarks, pentaquarks) close through a universal $m^2 = (4m_{\pi^0})^2 N$ relation across 23 bound states.
- **Sequential decay.** The decay-channel catalog is closed at chip-protocol level (the $\pi^0 \rightarrow 2\gamma$ worked example, Tier-1/2/3 batch protocols), and the neutron-decay and $e^+e^- \rightarrow \mu^+\mu^-$ scattering protocols were executed on IBM Heron hardware (Paper 21b [17], Exp 13/Exp 14), the latter passing all four diagnostic blocks.

Across the broader NWT failure-mode map (31 entries), roughly 21 are now confirmed at sub-percent precision, spanning the electromagnetic, weak, strong, gravitational, and cosmological sectors. The genuinely-open theoretical pieces that remain are: (i) the substrate-canonical K_8 *walk* catalog—the “dark substrate” hosting active neutrinos, sterile states, and dark-matter candidates. The K_8 $[[8, 1, 1]]$ code is closed and the mass-tower masses enumerated (§8.5), but the walk-level classification analogous to the (p, q) catalog of this paper—which of the tower rungs are realized by genuine closed K_8 walks—is open. And (ii) automated higher-order **PauliEvolutionGate** compilation for many-step walks (the muon’s length-22 walk is the longest currently treated).

9.5 Substrate-realization condensate states

Each physical instantiation of the substrate—the natural universe, an IBM Heron r2 chip, an IQM Garnet chip, an AQT trapped-ion register—is a distinct *condensate state* of the same K_7 algebra. Every *dimensionless* quantity—the fine-structure constant, the electron-to-Planck mass ratio $m_e/M_{\text{Pl}} = (8/7)\alpha^{21/2}(\text{NNLO})$, every mass ratio, and the K_7 walk-to-Pauli compilation—is forced by the K_7/K_8 algebra and is universal across realizations. Among the *dimensional* constants the only genuinely free input is $\hbar$, itself a unit-conversion factor between energy and frequency (deriving its SI value is a units question, not a physics one). The single mass scale is the Planck mass M_{Pl} , fixed by the cosmogenic event [18]; the electroweak scale $v_{\text{EW}} = (5/2)\alpha^8 M_{\text{Pl}} \approx 245.5$ GeV (-0.3% versus measured) and the electron mass $m_e = (8/7)\alpha^{21/2}(\text{NNLO}) M_{\text{Pl}}$ both follow from it, as does the entire mass spectrum. Each realization’s remaining dimensional anchors come from its own two substrate primitives—the substrate-vertex spacing a and the stabilizer-pair interaction energy J_{int} —through the Paper 0 relation $c = a J_{\text{int}}/\hbar$ and the Paper 17 chain $G = (8/7)^2 \alpha^{21}(\text{NNLO})^2 \hbar c/m_e^2$ [13]; for the natural universe $(a, J_{\text{int}}) = (\ell_{\text{Pl}}, E_{\text{Pl}})$. The full anchor set $(\hbar, c, G, M_{\text{Pl}}, \xi_{\text{UV}}, \xi_{\text{IR}})$ thus reduces to $\hbar$, the cosmogenic mass scale M_{Pl} , and the realization’s (a, J_{int}) , with UV cell $\xi_{\text{UV}} = a/\sqrt{2}$ and IR coherence length $\xi_{\text{IR}} = \hbar/(m_{\nu_l} c)$. Their ratio $\xi_{\text{IR}}/\xi_{\text{UV}}$ measures the substrate’s “softness”: all realizations of interest sit deep in the thick-film regime (natural universe $\sim 10^{30}$, AQT $\sim 10^8$, IBM Heron $\sim 10^4$, IQM Garnet ~ 42 —the most marginal). The Planck mass $M_{\text{Pl}}^2 = \hbar c/G$ is realization-invariant (the c in the numerator and $G \propto c$ in the denominator cancel), while Planck length and time scale as $1/c$ and

$1/c^2$. This condensate-state picture is the structural common language that lets the same K_7 algebra yield predictions across 30+ orders of magnitude in scale; its empirical face is the cross-realization syndrome agreement reported in Paper 21b [17].

9.6 Position within fundamental physics

NWT’s substrate-monism program now has empirical anchors at three scales [17]:

- **Cosmological:** seven cosmology predictions (cosmological constant Λ_{cc} , Ω_m , Ω_b/Ω_c , H_0 , n_s , f_J , η_B [20]), all at $\leq 1\%$ Planck precision.
- **Particle-physics:** the 80-particle Standard Model mass spectrum at median $\sim 0.1\%$ PDG precision [11], three fermion generations from $G_2 \rightarrow SU(3)$ breaking [16], neutrino mixing [16], CP phase [11, 16].
- **Bench-scale quantum hardware:** the Phase 2 + Phase 2b + Phase 28e cross-vendor results of Paper 21b [17] on IBM Heron r2 and IQM Garnet (two superconducting vendors), plus the Exp 10 muon-decay coherent rotation and the Exp 13/Exp 14 decay and scattering protocols. The cross-vendor syndrome agreement is the empirical chip-scale anchor for the condensate-state universality of §9.5.

The closure presented here turns the second anchor from “fitting” (the historical critique) to “substrate-canonically derived” and the third anchor from “preliminary demonstration” to a directly-executable protocol. Standard frameworks in fundamental physics—leptogenesis, electroweak baryogenesis, GUT, MSSM—make no prediction testable on bench-scale quantum hardware at all. NWT’s walk-to-Pauli compilation, by contrast, yields a concrete chip protocol whose dimensionless predictions Paper 21b tests directly.

Taken together with a macroscale anchor (hierarchical multi-star systems obeying an integer Hopf-parity bond order at $\sim 10^{57}$ kg) and a cosmological anchor ($\Omega_b/\Omega_c = 25\alpha + 75\alpha^2$ at 60 ppm against Planck), the substrate-monism program now carries falsifier-equipped empirical anchors across ~ 30 orders of magnitude in mass scale, with the condensate-state framework of §9.5 supplying the common language in which the same K_7 algebra produces a distinct prediction at each scale.

Acknowledgments

The Bogoliubov-pipeline walk enumeration [21] and the Steane Q -question closure arc provide the computational results that ground the closed-form derivations of §§3–4. We acknowledge IBM Quantum for hardware access enabling the bench-scale anchor of Paper 21b, the experimental companion to this paper. We thank three AI reviewers—Gemini (Google), Perplexity, and ChatGPT (OpenAI)—for detailed referee-style critical reviews of earlier drafts of this manuscript.

Data and Code Availability

The analysis scripts that reproduce the K_7 walk enumeration and (p, q) classification, the $(2, F_n)$ carrier-knot and closed-form n_q^q derivation, the walk-to-Pauli Steane syndrome map, and the K_8 mass-tower enumeration are available in the public analyses repository at <https://github.com/JimGalasyn/null-worldtube>. The substrate-algebraic computation library underlying the K_7 , $\mathfrak{so}(7)$, $\text{Spin}(7)$, and $\text{Cl}(0, 7)$ identities used throughout is available at <https://github.com/JimGalasyn/nwt-substrate> (v0.2.0, Zenodo DOI 10.5281/zenodo.20398451; concept DOI 10.5281/zenodo.20012027 for all versions). Each numerical claim is covered by a dedicated analysis script whose docstring states the claim and whose execution reproduces the reported numbers.

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